DASC 3240, L05

Overview + Plotting one variable

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Today's outline

- Package overview
  1. Matplotlib
  2. Seaborn
  3. ggplot2 & plotnine
  4. Plotly
- Plotting one variable
  - Histogram and density plot

Overview of data visualization packages

Do we need to rely on packages?

There are many packages for data visualization. There are pros and cons of using packages for data visualization.

Pros:

- Packages often provide a wide range of visualization options and customization features that can **save time and effort compared to writing code from scratch**.
- Packages are usually well-documented and have an active community of developers and users who can **provide support and guidance**.
- Packages are often **optimized for performance** and can handle large datasets more efficiently than custom code.

Do we need to rely on packages?

There are many packages for data visualization. There are pros and cons of using packages for data visualization.

Cons:

- Packages can **sometime be limited in terms of flexibility**, and may not be able to meet specific or unique visualization needs.
- **Using packages without fully understanding how they work can lead to errors**. Users have to invest time and effort in learning how to use packages.
- **Packages can have specific dependencies and versions**. If packages and their dependencies are frequently updated, you might not able to reproduce your figures in near future. <= **An important issue in terms of reproducibility**

Do we need to rely on packages?

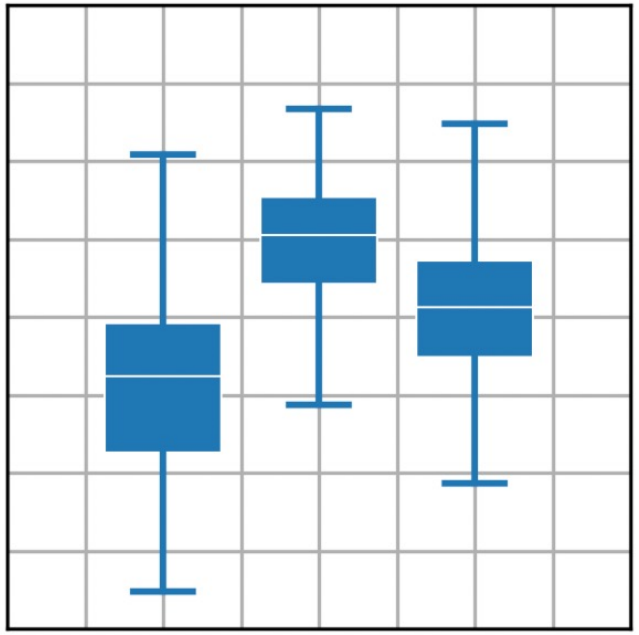
- Overall, it is **a good idea** to use packages to visualize your data efficiently and elegantly. However, **if you rely on several packages, it is important to record the package versions** (Python version, too).
- Otherwise, **your code might not work in near future and cannot reproduce the figures that you created previously**.

Matplotlib

Matplotlib is a comprehensive library for creating static, animated, and interactive visualizations in Python. Matplotlib makes easy things easy and hard things possible.

Matplotlib has large user communities.

matplotlib Plot types User guide Tutorials Examples

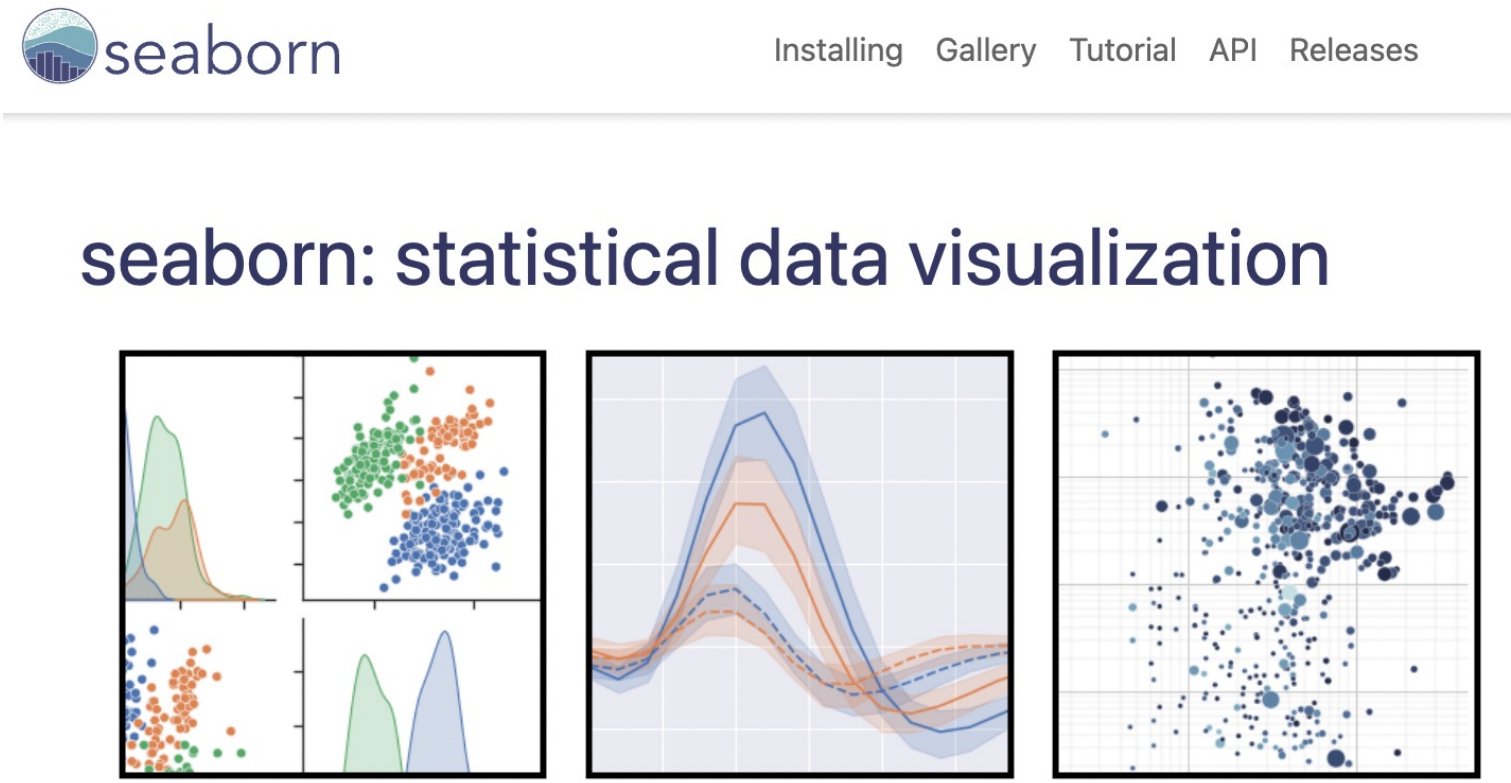


boxplot(X)

Seaborn

Seaborn is a Python data visualization library based on **matplotlib**. It provides a high-level interface for drawing attractive and informative statistical graphics.

It requires less amount of code compared with matplotlib.



ggplot2 & plotnine

ggplot2 is an R package (not Python) and is a system for declaratively creating graphics, based on *The Grammar of Graphics*.

plotnine is an implementation of the grammar of graphics in Python based on **ggplot2**.

ggplot2 3.4.4 Reference News Articles Extensions

ggplot2

plotnine 0.12.4 API Gallery Tutorials Site Page

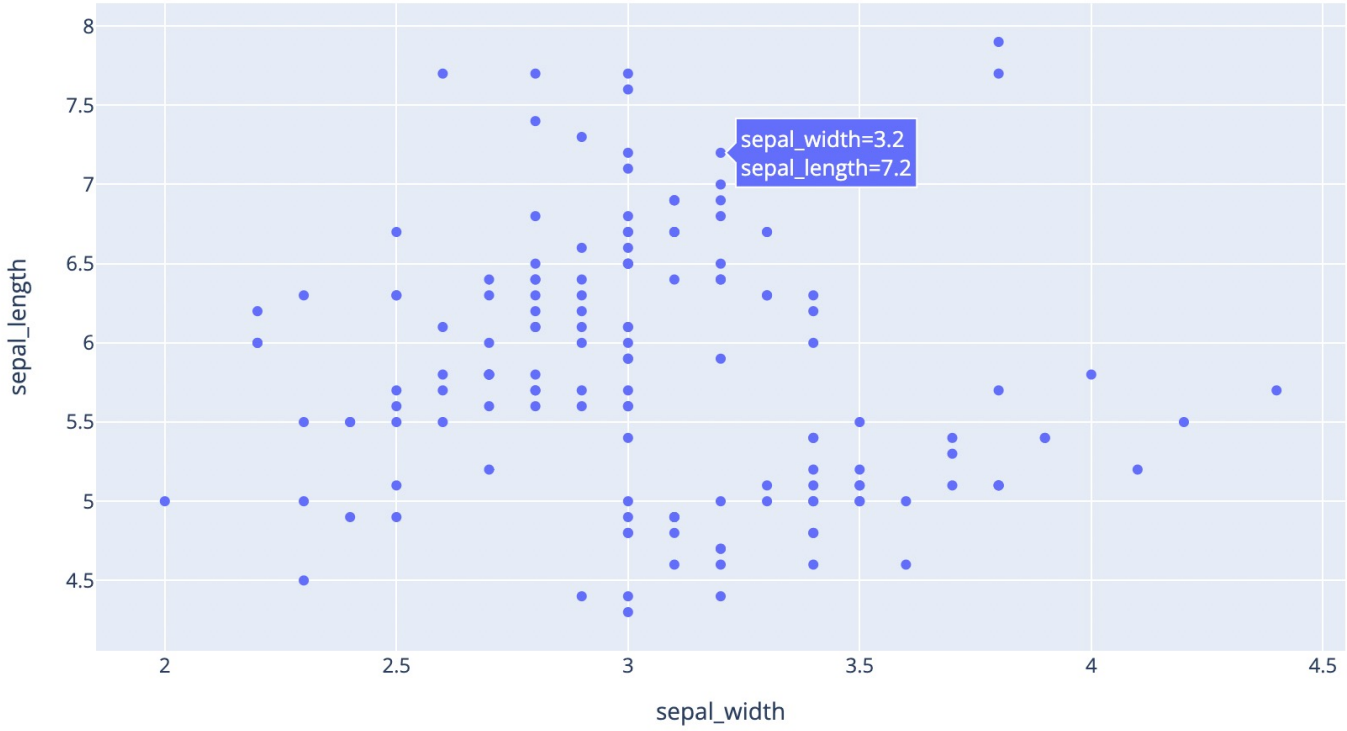
A Grammar of Graphics for Python

https://ggplot2.tidyverse.org/ https://plotnine.readthedocs.io/en/v0.12.4/

plotly

plotly.py is an interactive, open-source, and browser-based graphing library for Python.

plotly can be used from other languages, for example **R**, **Matlab**, **Julia**, ...



https://plotly.com/python/line-and-scatter/

Which package(s) should we use?

- **Basic figures can be generated by most packages.**
- **There are differences, for example, in a default setting and syntax.** A different person has a different preference.
- In this lecture, we will use multiple packages. **Once you get familiar, you can choose one.**
- Plotly is suitable for "Interactive figures."
- **NOTE**: There packages are being actively developed and their functionalities may change in near future. It is a good idea to check these packages sometimes (once a year?).

Plotting one variable

HISTOGRAM AND DENSITY PLOT

What should we consider when we visualize data?

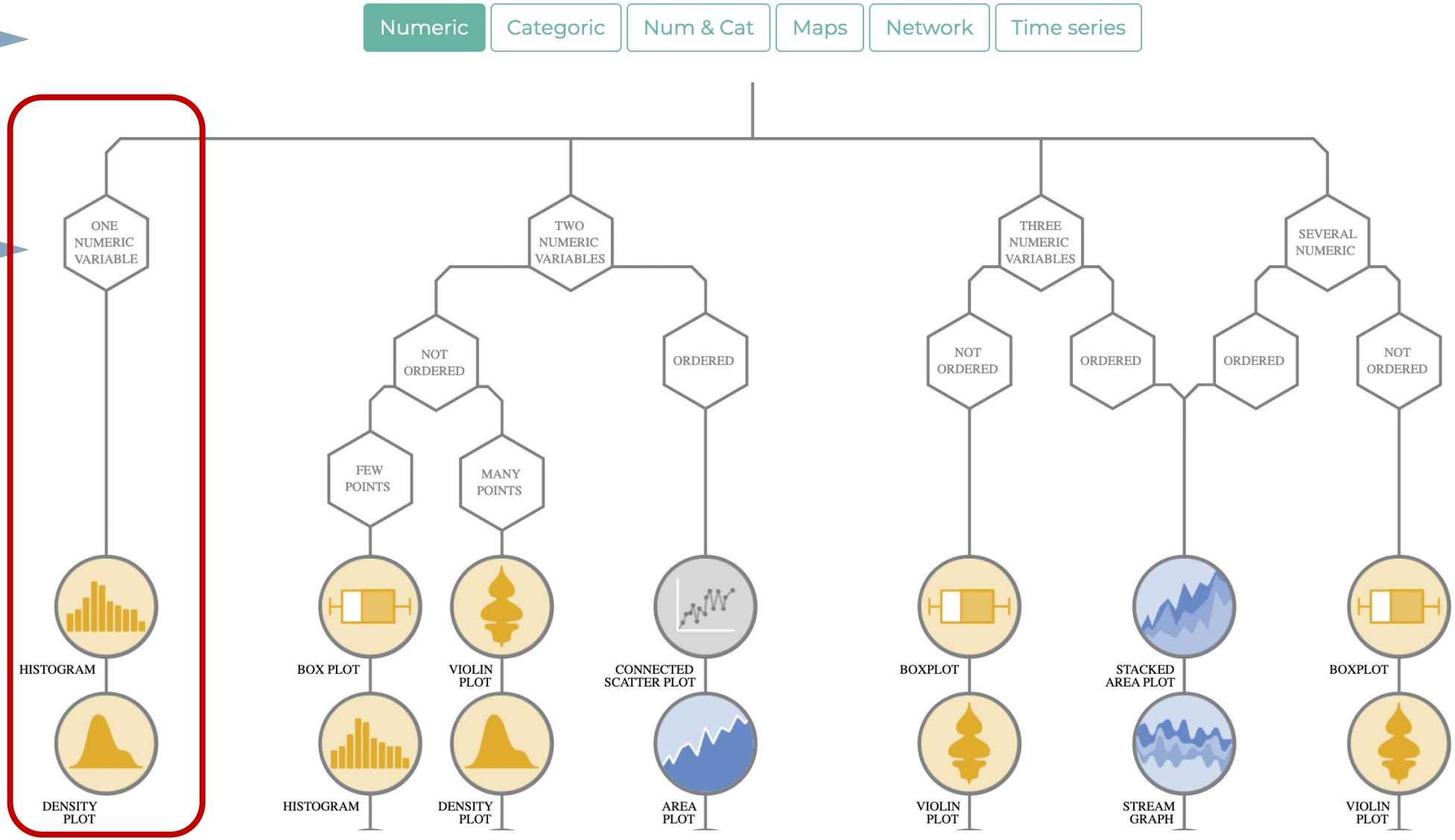
1. **The number of variable**
2. **Is the variable(s) numeric, or categorical?**

(there are other factors that we should consider)

How many variables? Numeric or categorical?

Numeric or categorical?

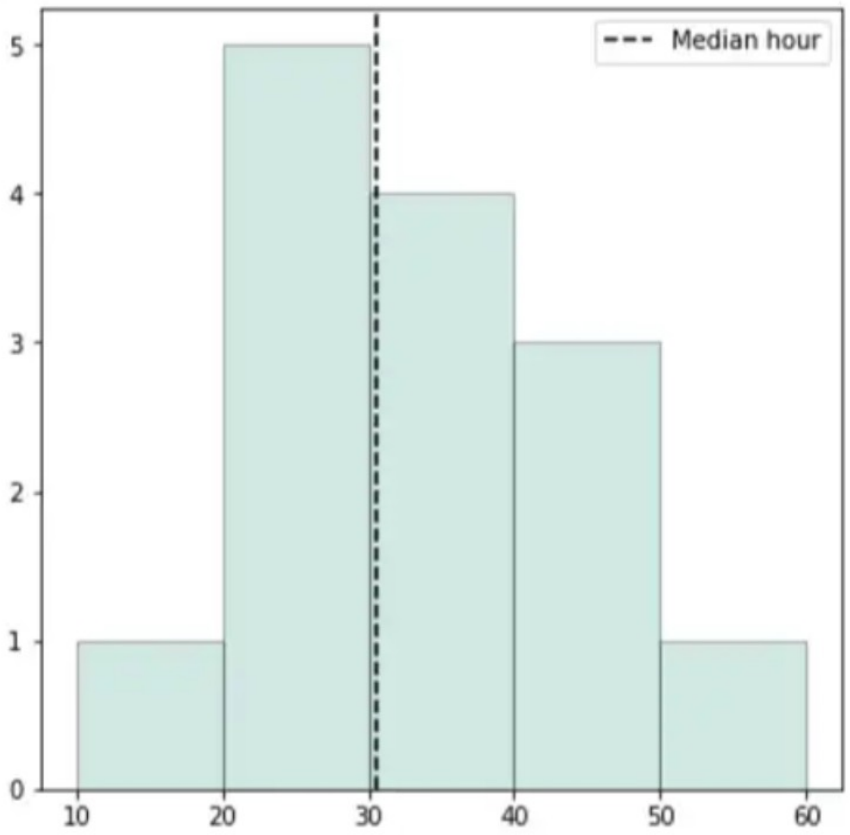
The number of variables



https://www.data-to-viz.com/

Histogram

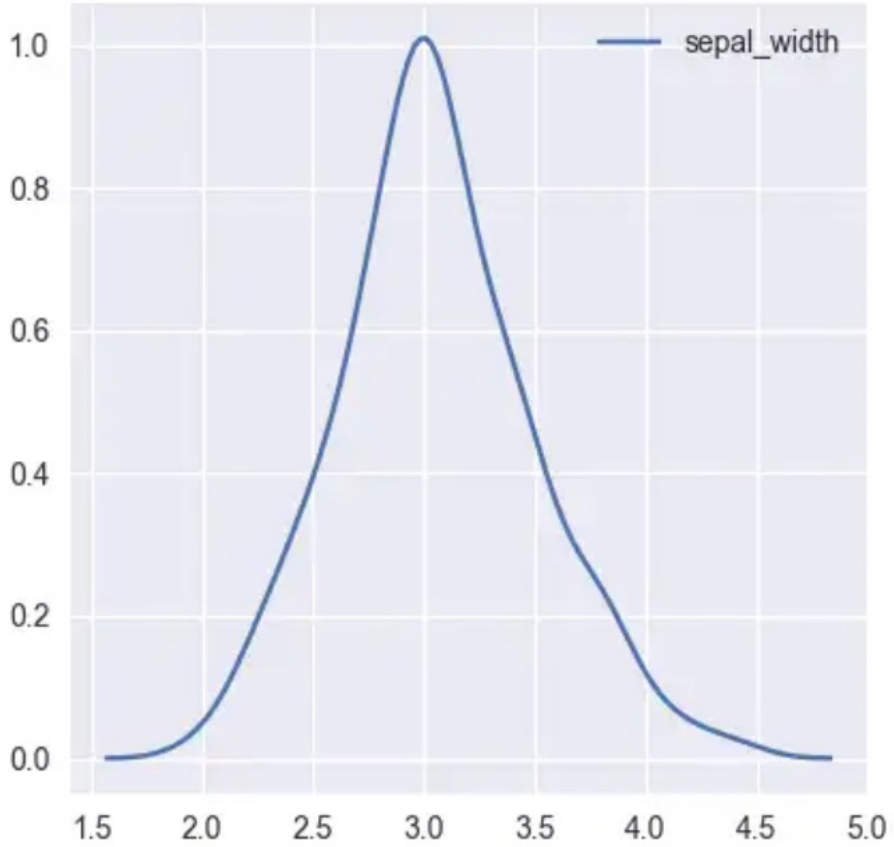
- A **Histogram** represents the distribution of a **numeric** variable for **one** (or several) groups.
- The values are split in **bins;** each bin is represented as a bar.
- We will draw histograms using **seaborn** and **matplotlib.**
- Examples start with very simple, beginner-friendly histograms and progressively increase in complexity.



https://python-graph-gallery.com/histogram/

Density plot

- **Density plots** allow you to visualize the distribution of a **numeric** variable for **one** (or several) group(s).
- Note that two approaches exist to build them in python: the first one consists of computing a kernel density estimate, and the second one involves building a high resolution histogram.



https://python-graph-gallery.com/density-plot/

Let’s try

https://github.com/ong8181/DASC3240

Open Jupyter notebook in your local PC

If Jupyter Notebook is already installed in your PC, the most straightforward way to run the scripts is probably downloading all the files in this repository and open them using jupyter notebook. Alternatively, you can open them in Google Colab, but reading demo data is a bit tricky.

Open Jupyter Notebook in Google Colab

Materials for lectures

- Open in Colab L04_ReviewPython.ipynb
- Open in Colab L05_OneVariable.ipynb
- Open in Colab L06_OneVariable-HandsOn.ipynb

Iris

https://vincentarelbundock.github.io/Rdatasets/doc/datasets/iris.html
https://vincentarelbundock.github.io/Rdatasets/csv/datasets/iris.csv

Edgar Anderson's Iris Data

Description

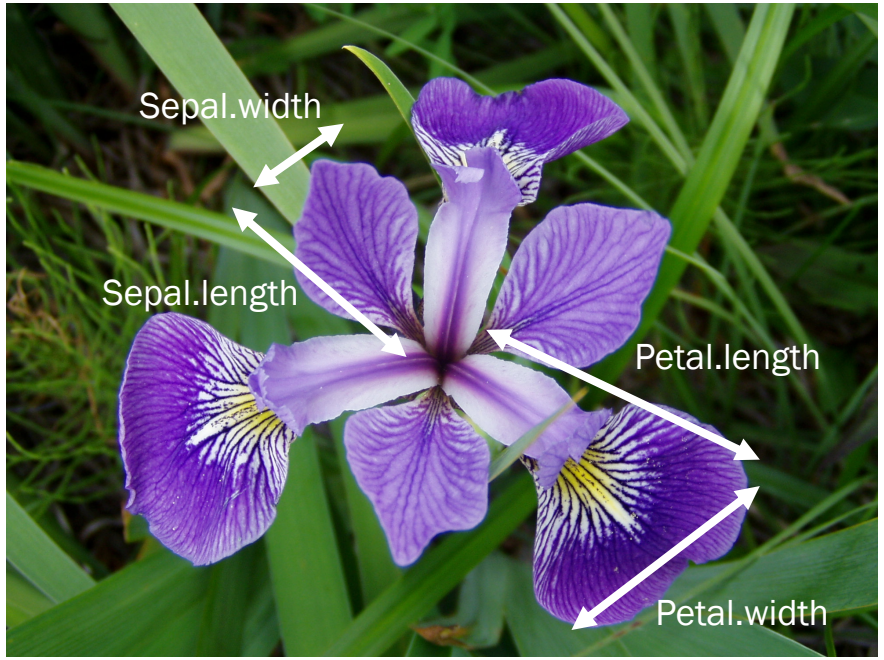
This famous (Fisher's or Anderson's) iris data set gives the measurements in centimeters of the variables sepal length and width and petal length and width, respectively, for 50 flowers from each of 3 species of iris. The species are *Iris setosa*, *versicolor*, and *virginica*.

https://en.wikipedia.org/wiki/Iris_flower_data_set#/media/File:Iris_versicolor_3.jpg

Iris

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3.0	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
4	4.6	3.1	1.5	0.2	setosa
5	5.0	3.6	1.4	0.2	setosa
6	5.4	3.9	1.7	0.4	setosa

Sepal.Length	Sepal length
Sepal.Width	Sepal width
Petal.Length	Petal length
Petal.Width	Petal width
Species	Iris species (3 species)



https://en.wikipedia.org/wiki/Iris_flower_data_set#/media/File:Iris_versicolor_3.jpg